

CS & IT ENGINEERING



Data Structure & Programming

Trees and Graphs

Lecture No.- 04



By- Satya sir

Recap of Previous Lecture



- Binary Search Tree

- Insertion / Construction

- Deletion

- Tree Traversal Practice

- Theorems on Binary Trees



Topics to be Covered



Binary Heap

- Nature
- Array Representation of Binary Tree
- Insertion into Binary Heap





Topic : Binary Search Tree

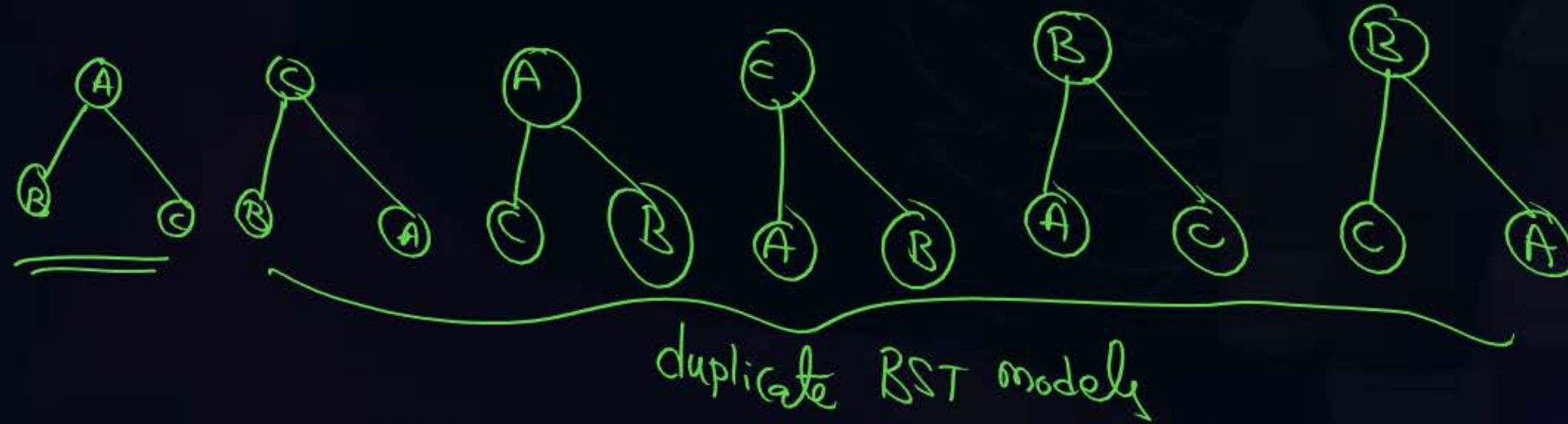


How many distinct binary search trees can be created out of 4 distinct keys?

- (A) 5
- ~~(B) 14~~
- (C) 24
- (D) 42

$$n_{c_n} = \frac{n!}{(n-r)! r!}$$
$$\frac{8!}{(8-4)! * 4!} = \frac{8!}{4! 4!}$$
$$\frac{2n_{c_n}}{n+1}$$
$$n=4 \Rightarrow \frac{8c_4}{5} = \frac{8 \times 7 \times 6 \times 5 \times 4!}{4! \times 4!} * \frac{1}{5} = 14$$

GATE 2005





Topic : Binary Heap - 1

A Node at index 'x' will have, it's Parent Node at index $\lfloor \frac{x-1}{2} \rfloor$

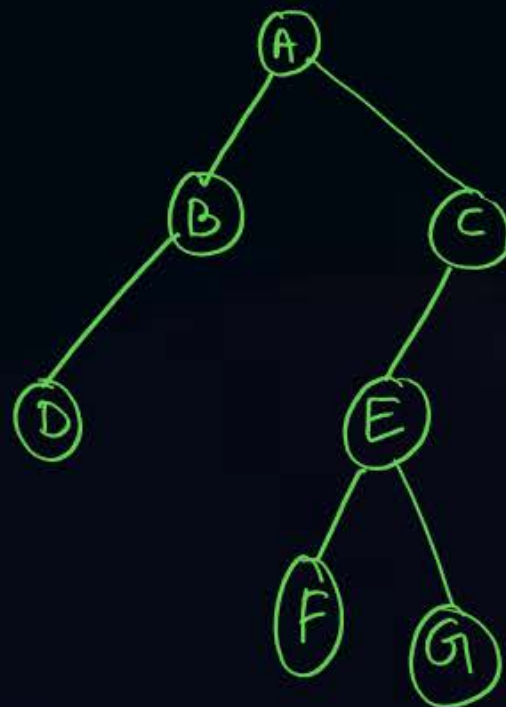


Array Representation of a Binary Tree

$$2i+1 = x \\ \Rightarrow i = \frac{x-1}{2}$$

$$2i+2 = x \\ i = \frac{x-2}{2}$$

Ex:1



Parent Node index

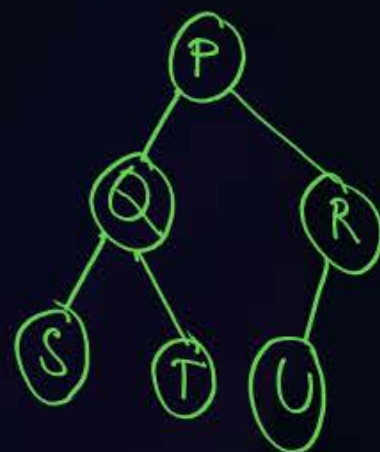
i = 0 $2i+1$ $2i+2$

1 2 3 4 5 6 7 8 9 10 11 12

A	B	C	D	—	E	—	—	—	—	—	F	G					
	$\uparrow$	$\uparrow$									$\uparrow$	$\uparrow$					
	$\lfloor \frac{1-1}{2} \rfloor$	$\lfloor \frac{2-1}{2} \rfloor$									Left	Right					
														Left	Right	Left	Right

Node at index i, will have it's left child at index = $2i+1$
Right child at index = $2i+2$

Ex:2



0	1	2	3	4	5	6	7	8	9	10	11	12
P	Q	R	S	T	U							

← No Space B/w Elements → Right ⇒ CBT



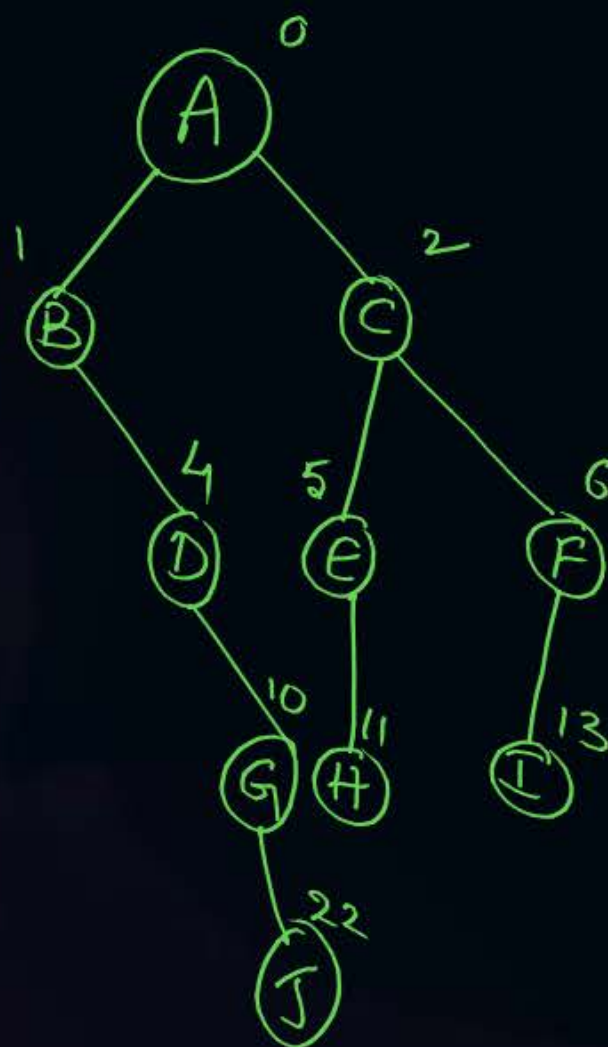
Topic : Binary Heap - 1



Given Array Representation.

0	1	2	3	4	5	6	7	8	9	10	11	12	13	21	22	
A	B	C	-	D	E	F	-	-	-	G	H	-	I	...	J	...

$$\text{Parent index} = \left\lfloor \frac{22-1}{2} \right\rfloor = \left\lfloor \frac{21}{2} \right\rfloor = 10$$





Topic : Binary Heap - 1



Binary Heap

A Binary Tree, that satisfy 2 Properties :

① Complete Binary Tree $\Rightarrow$ Shape Property / Structuring Property

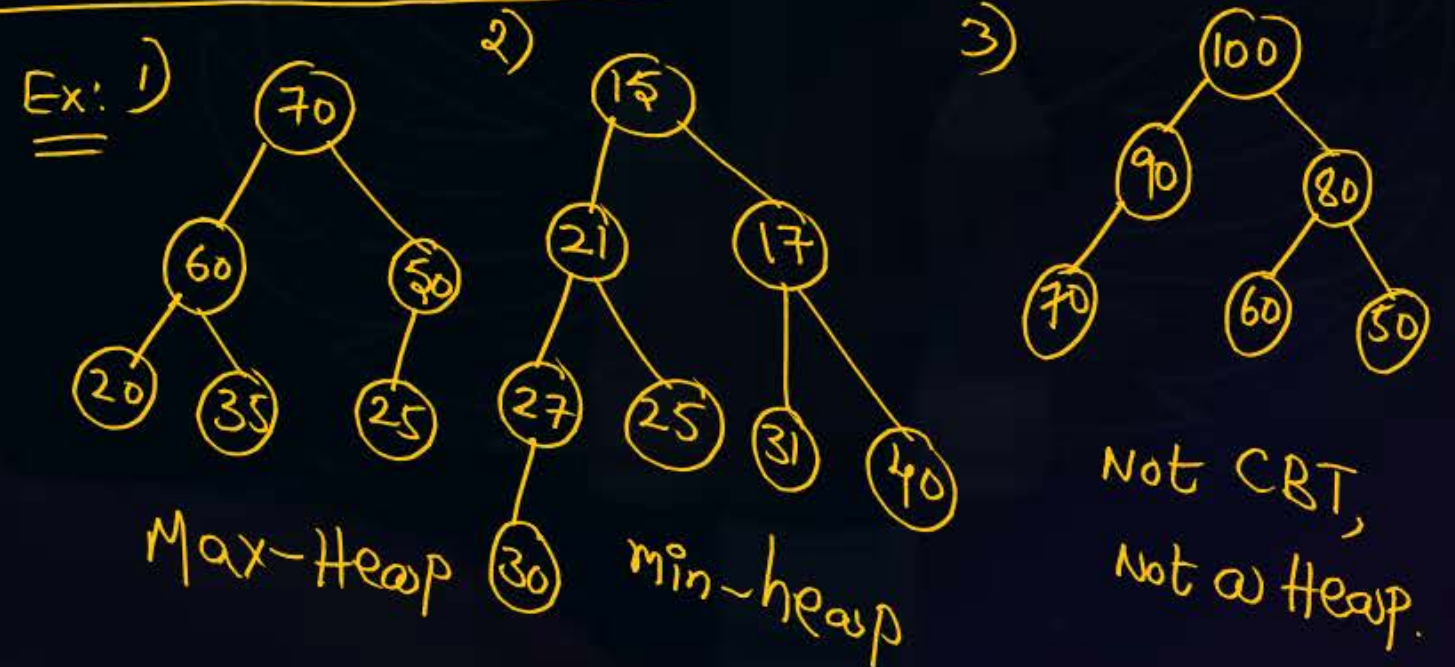
② Every Parent Value $>$ All children values (or) Every Parent value $<$ All children values

Ordering Property (or) heap Property

2 Types of Binary Heap :

① Max-Heap : CBT, Parent $>$ children

② min-heap : CBT, Parent $<$ children





Topic : Binary Heap - 1



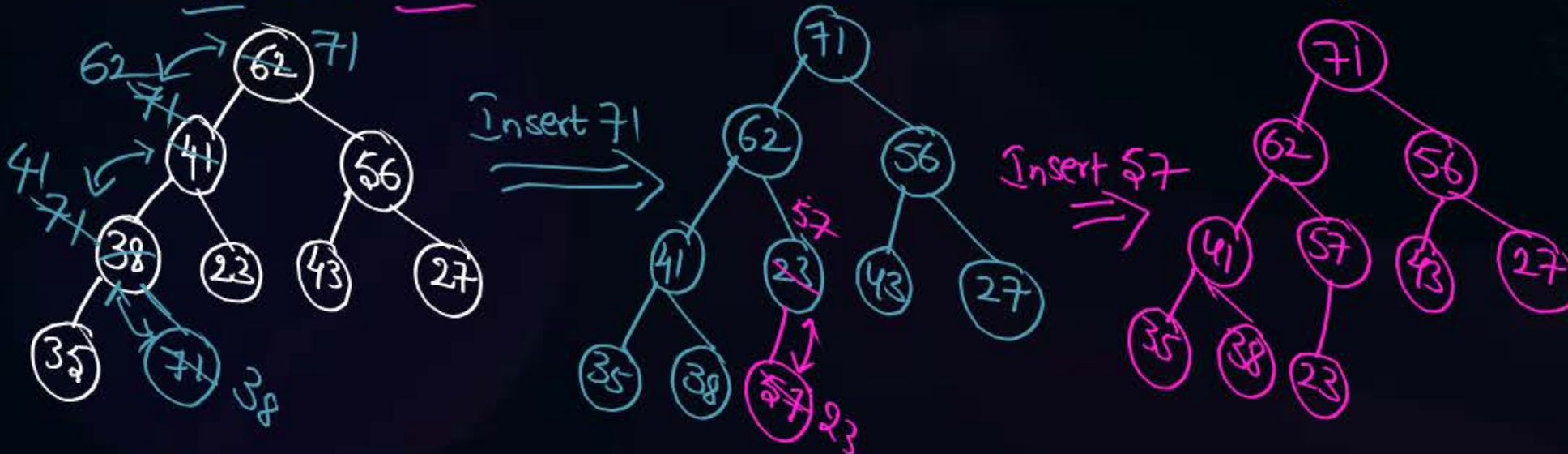
Construction of Heap : Insertion of multiple nodes one after another.

Insertion into Binary Heap

① Insert New Node at End, as last leaf node.

② Heapify $\left\{ \begin{array}{l} \rightarrow \text{For Max-Heap, if } \underline{\text{Parent} < \text{New Node}}, \text{ swap;} \\ \rightarrow \text{For min-heap, if } \underline{\text{Parent} > \text{New Node}}, \text{ swap;} \end{array} \right\}$ Repeat this until Complete Tree is Heapified.

Ex: Consider a Binary Heap with Elements 62, 41, 56, 38, 23, 43, 27, 35. After insertion of New Nodes, 71 and 57 in that order, The Resultant Binary Heap will be 71, 62, 56, 41, 57, 43, 27, 35, 38, 23

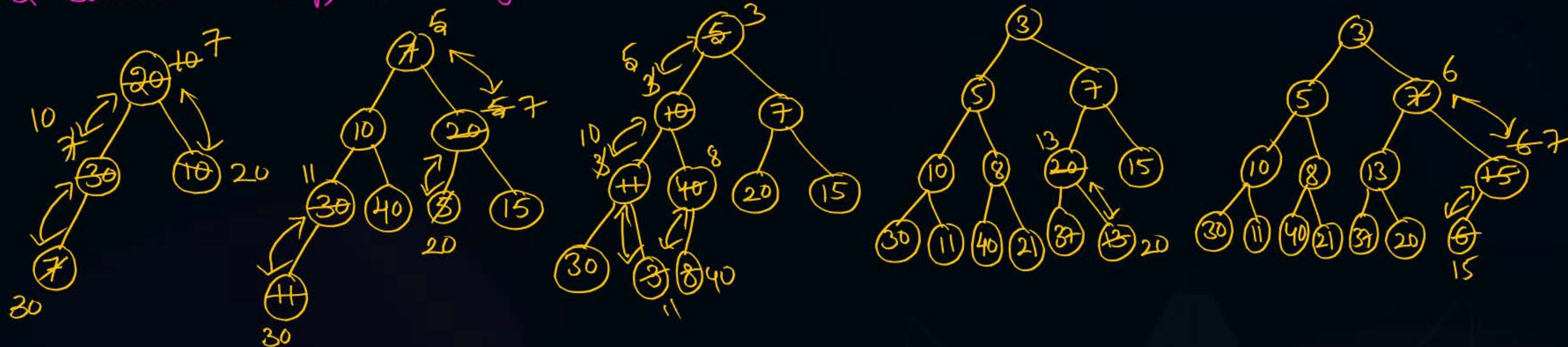




Topic : Binary Heap - 1



#Q. Construct min-heap, by inserting elements in the order : 20, 30, 10, 7, 40, 5, 15, 11, 3, 8, 21, 37, 13, 6.



Resultant min-heap: 3, 5, 6, 10, 8, 13, 7, 30, 11, 40, 21, 37, 20, 15



Topic : Binary Heap - 1



H/W

Construct Max-Heap, by inserting elements in the order: 6, 7, 9, 11, 13, 20, 25, 27, 31, 42, 56.



2 mins Summary



- Array Representation of Binary Tree

- Binary Heap

- Insertion

- Construction

To be contd... A simple hand-drawn smiley face with two dots for eyes and a curved line for a mouth.



THANK - YOU